



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AMEERIA SERVICE STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

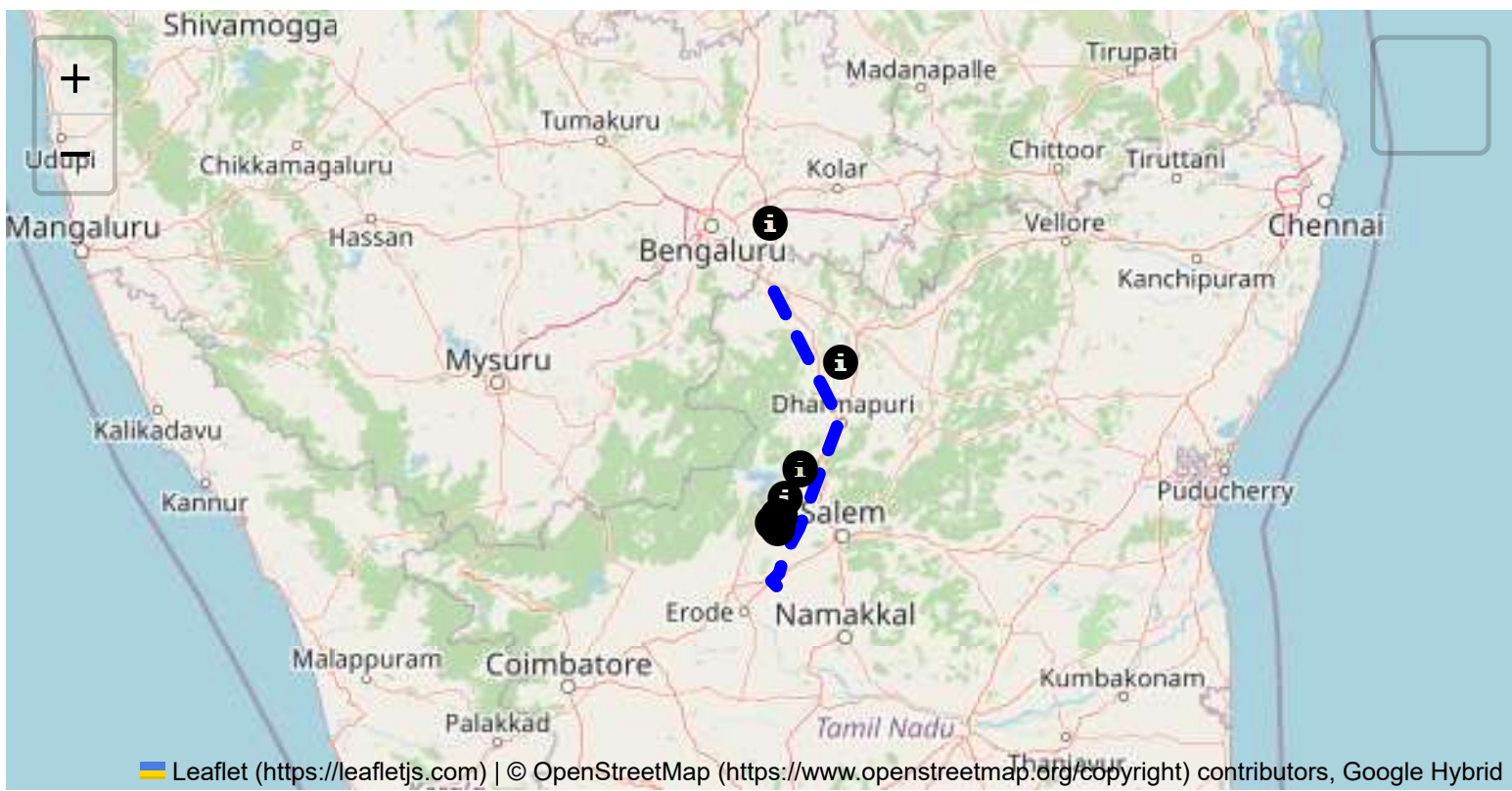
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

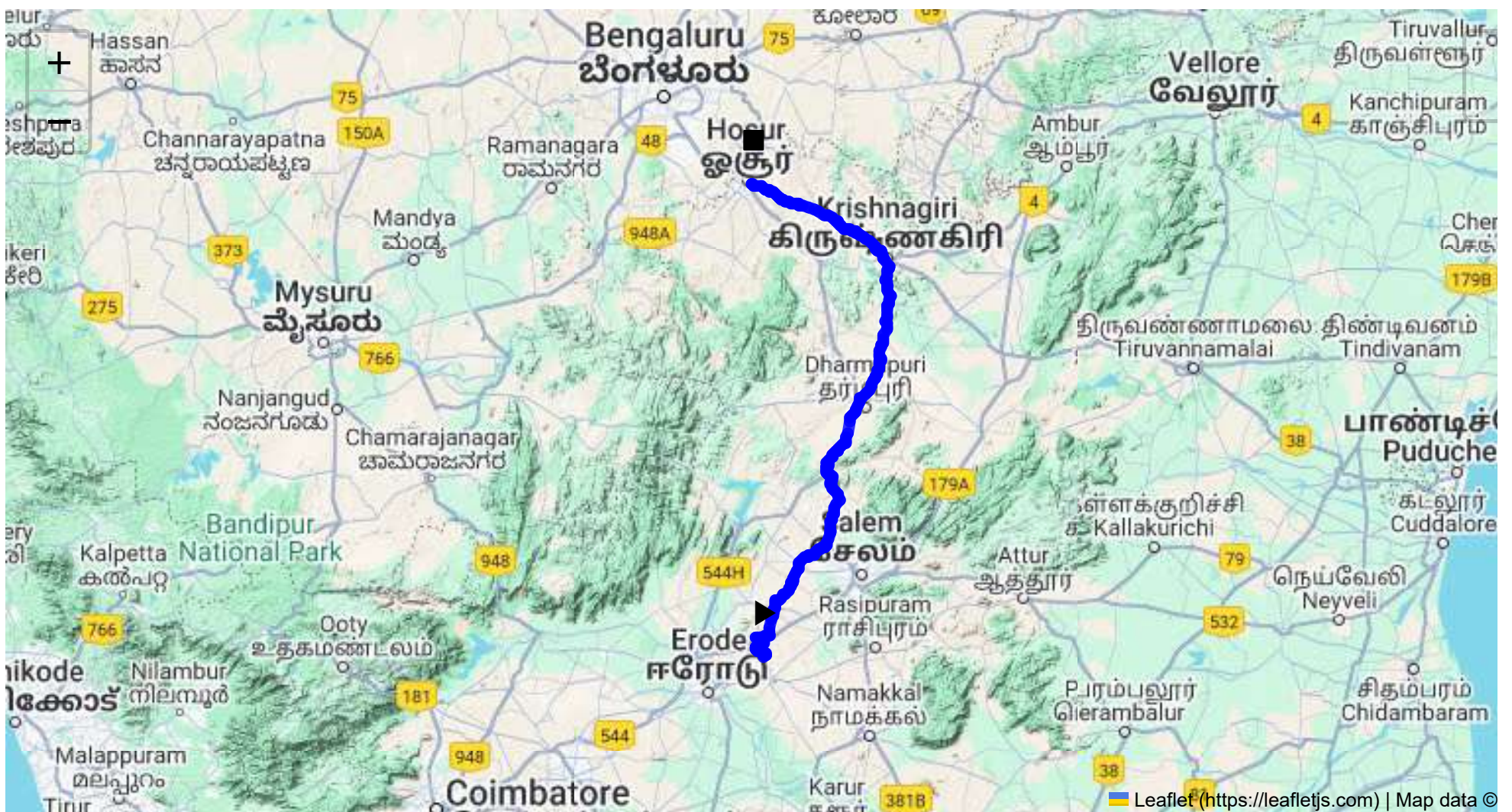
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 198.82 km
Estimated Duration: 3.9 hours
Adjusted Duration (Heavy Vehicle): 4.8 hours
Start: (11.4381, 77.8734)
End: (12.732447, 77.835465)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

This route connects Sangagiri to Hosur in Tamil Nadu, spanning approximately 198.82 kilometers and includes several state highways and local roads. Key stops or transit areas include Sankari RS, Sankari West, VN Palayam, Ayveli, Konganapuram, Desavilakku, Tharamangalam, and A.Reddihalli. It's a blend of urban, semi-urban, and rural roads.

2. Typical Weather Conditions and Potential Weather-related Hazards

Tamil Nadu typically experiences a tropical climate with the following:

- **Summer (March to June):** High temperatures reaching up to 40°C, potential for road surface damage.
- **Monsoon (July to September):** Heavy rains may lead to flooding, especially in low-lying areas. Reduced visibility and slippery roads are common.
- **Winter (November to February):** Mild and generally dry weather, but morning fog might be encountered.

3. Analysis of Traffic Patterns

- **Peak Hours:** Generally, morning between 8:00-10:00 AM and evening 5:00-8:00 PM face the highest traffic due to commuting.
- **Congestion-prone Areas:** Urban centers like Sangagiri, Sankari, and towns along SH 86 may experience slow traffic.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Varies greatly; state highways such as SH 86 are better maintained compared to smaller local roads that may have potholes or uneven surfaces.
- **Infrastructure:** Limited emergency facilities on local roads, few truck stops, gas stations, and repair shops particularly after A.Reddihalli towards the more rural stretches.

5. Suggestions for Alternative Routes for Emergencies

- In case of road blocks or extreme weather, alternative routes using NH44 can be used, which although longer, provide better infrastructure and facilities.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Transport of hazardous materials requires specific permits.
- **Restricted Areas:** Must avoid densely populated areas and follow designated routes for hazardous materials.

7. Overview of Historical Incidents

- **Heavy Vehicles and Hazardous Materials:** Historical records show occasional incidents involving chemical transport owing to improper handling or inadequate vehicle maintenance.

8. Environmental Considerations and Sensitive Areas

- Proximity to wildlife areas and agricultural land means strict adherence to environmental protection guidelines is necessary.
- Avoid spillage or any emissions that could affect local flora and fauna.

9. Analysis of Communication Coverage

- Generally good near urban centers but potential dead zones in rural areas, particularly between Devannagoundanur and A.Reddihalli.

10. Estimated Emergency Response Times

- **Urban Areas:** 30-45 minutes due to better access.
- **Rural Segments:** Can exceed 1 hour owing to distance and accessibility challenges.

11. Overall Summary of Risk Assessment

This route presents generally moderate risk due to varied road quality and the potential for weather-related hazards. Congestion in urbanized areas and limited emergency response in rural sections raise concerns. Careful driving, route planning around weather patterns, and adherence to local regulations can mitigate most risks. Communication plans and alternative routes should be established for emergency situations. Transporters should remain vigilant, especially when traversing sensitive environmental areas and adhere to strict compliance regarding hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.4629262, 77.84776280000001	10 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	Medium	11.43995, 77.87431	30 KM/Hr
3	Turn	High	11.44107, 77.87528	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Blind Spot	Blind Spot	11.46304, 77.84810	10 KM/Hr
8	Turn	Medium	11.47247, 77.86225	30 KM/Hr
9	Turn	Medium	11.47358, 77.86766	30 KM/Hr
10	Turn	Medium	11.47487, 77.86913	30 KM/Hr
11	Turn	High	11.48237, 77.87241	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.49297, 77.86810	15 KM/Hr
13	Blind Spot	Blind Spot	11.49012, 77.85118	10 KM/Hr
14	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
15	Turn	High	11.49469, 77.88488	15 KM/Hr
16	Turn	Medium	11.57662, 77.90612	30 KM/Hr
17	Turn	High	11.59235, 77.90675	15 KM/Hr
18	Turn	Medium	11.63843, 77.94821	30 KM/Hr
19	Turn	High	11.64301, 77.95190	15 KM/Hr
20	Turn	Medium	11.64747, 77.95129	30 KM/Hr
21	Turn	Medium	11.69642, 77.97213	30 KM/Hr
22	Turn	Medium	11.71402, 77.99586	30 KM/Hr
23	Turn	Medium	11.71704, 77.99948	30 KM/Hr
24	Turn	Medium	11.71918, 78.01127	30 KM/Hr
25	Turn	Medium	11.72708, 78.02197	30 KM/Hr
26	Turn	High	11.73773, 78.03459	15 KM/Hr
27	Turn	High	11.73651, 78.03873	15 KM/Hr
28	Turn	Medium	11.74170, 78.04640	30 KM/Hr
29	Turn	High	11.86860, 78.08371	15 KM/Hr
30	Turn	Medium	11.87954, 78.08191	30 KM/Hr
31	Turn	Medium	11.91414, 78.05888	30 KM/Hr
32	Turn	Medium	11.92597, 78.06510	30 KM/Hr
33	Turn	Medium	11.93138, 78.06487	30 KM/Hr
34	Turn	Medium	11.93451, 78.06300	30 KM/Hr
35	Turn	Medium	11.93710, 78.05390	30 KM/Hr
36	Turn	Medium	11.95430, 78.04683	30 KM/Hr
37	Turn	Medium	11.95668, 78.04785	30 KM/Hr
38	Turn	High	11.95824, 78.04981	15 KM/Hr
39	Turn	Medium	11.95992, 78.04947	30 KM/Hr
40	Turn	High	11.96401, 78.05134	15 KM/Hr
41	Turn	High	11.96504, 78.04975	15 KM/Hr
42	Turn	Medium	11.96679, 78.04942	30 KM/Hr
43	Turn	Medium	12.16983, 78.16769	30 KM/Hr
44	Turn	Medium	12.19881, 78.18157	30 KM/Hr
45	Turn	Medium	12.22459, 78.19136	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Turn	Medium	12.24731, 78.19601	30 KM/Hr
47	Turn	Medium	12.25060, 78.19977	30 KM/Hr
48	Turn	Medium	12.25329, 78.20039	30 KM/Hr
49	Turn	Medium	12.29110, 78.19909	30 KM/Hr
50	Turn	Medium	12.32082, 78.20751	30 KM/Hr
51	Turn	Medium	12.39904, 78.21764	30 KM/Hr
52	Turn	Medium	12.40156, 78.21845	30 KM/Hr
53	Turn	Medium	12.40589, 78.22275	30 KM/Hr
54	Turn	Medium	12.42428, 78.22671	30 KM/Hr
55	Turn	Medium	12.44389, 78.21824	30 KM/Hr
56	Turn	Medium	12.50166, 78.22333	30 KM/Hr
57	Turn	Medium	12.50553, 78.22318	30 KM/Hr
58	Turn	Medium	12.59749, 78.13593	30 KM/Hr
59	Turn	Medium	12.63696, 78.07061	30 KM/Hr
60	Turn	Medium	12.67143, 77.97400	30 KM/Hr
61	Turn	Medium	12.71181, 77.89359	30 KM/Hr
62	Turn	Medium	12.71126, 77.88964	30 KM/Hr
63	Turn	Medium	12.73230, 77.83664	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Lakshmi Hospital	11.4723746, 77.8646891	30 km/h	Medium
1	hospital	ESI Hospital	11.4717067, 77.8642668	30 km/h	Medium
2	clinic	Sutha Clinic	11.473718, 77.862614	30 km/h	Medium
3	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
4	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
5	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
6	hospital	Ck Hospital	11.473504, 77.8650523	30 km/h	Medium
7	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
8	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
11	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
13	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
14	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
15	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
16	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
17	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
18	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
19	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
20	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
21	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
22	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
24	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
25	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
26	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
27	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
28	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
29	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
30	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
31	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
32	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
33	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
34	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
35	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
37	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
38	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
39	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
43	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
44	clinic	Shri Narayana Bone & Joint Clinic	12.523617, 78.210613	30 km/h	Medium
45	clinic	AZ Family Clinic	12.5234942, 78.2127422	30 km/h	Medium
47	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
48	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
49	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium
54	pharmacy	Mithran Medicals	12.6837657, 77.93838	30 km/h	Medium
55	pharmacy	Jaisheka Medical	12.7116723, 77.890613	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
57	pharmacy	H2 Pharmacy	12.7301802, 77.8401037	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
9	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
10	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
23	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
40	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
41	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
42	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
46	school	Vivekananda School	12.5231937, 78.2121429	30 km/h	Medium
50	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium
51	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium
52	college	Jeeva College of Nursing	12.6067058, 78.1191944	30 km/h	Medium
53	school	New Little Flower Matriculation Higher Secondary School	12.6607923, 78.0169012	30 km/h	Medium
56	college	Adhiyamaan College of Engineering	12.7152133, 77.8678715	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.4629262, 77.84776280000001



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44107, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46304, 77.84810



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47247, 77.86225



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47358, 77.86766



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47487, 77.86913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.48237, 77.87241



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49297, 77.86810



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49012, 77.85118



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57662, 77.90612



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59235, 77.90675



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63843, 77.94821



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64301, 77.95190



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64747, 77.95129



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69642, 77.97213



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71402, 77.99586



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71704, 77.99948



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71918, 78.01127



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72708, 78.02197



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73773, 78.03459



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03873



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74170, 78.04640



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.86860, 78.08371



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87954, 78.08191



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91414, 78.05888



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92597, 78.06510



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93138, 78.06487



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93451, 78.06300



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93710, 78.05390



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95430, 78.04683



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95668, 78.04785



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95824, 78.04981



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95992, 78.04947



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96679, 78.04942



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.16983, 78.16769



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24731, 78.19601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25060, 78.19977



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32082, 78.20751



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39904, 78.21764



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40156, 78.21845



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40589, 78.22275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42428, 78.22671



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.44389, 78.21824



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50166, 78.22333



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50553, 78.22318



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.59749, 78.13593



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.63696, 78.07061



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.67143, 77.97400



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71126, 77.88964



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.73230, 77.83664
